

Course Code: IBU 060	FUNDAMENTALS OF PROGRAMMING		
Level: Undergraduate	Year: I	Semester: I	ECTS Credits: 8
Status: Mandatory / Elective	Hours / Week: 3+2	Total Hours: 45+30=75	
Course Description	This course is designed to be an introduction to the fundamentals of programming. Students will design, write, and debug computer programs. No previous knowledge of programming is required.		
Course Objectives	- Introduce students to the intermediate concept of programming. - Introduce students to basic algorithm design principles. - Substantially strengthen students’ programming skills by requiring them to program a large number of small but challenging problems continuously. - Encourage them to strive for excellence by introducing them to a competitive environment where part of their performance will be based on the performance of their peers.		
Course Content	Week 1: Introduction (anaconda, venv, jupiter)		
	Week 2: Variables and Data Types		
	Week 3: Conditionals and Iterations		
	Week 4: Functions		
	Week 5: Strings		
	Week 6: Lambda		
	Week 7: Modules		
	Week 8: Recursion		
	Week 9: Lists		
	Week 10: Dictionaries		
	Week 11: Sets and Tuples		
	Week 12: Files (Binary and Text)		
	Week 13: Exceptions		
	Week 14: Python outside of IDE - numpy		
	Week 15: Wrap up		
Teaching Methods Description	Interactive lectures and communication with students Discussions and group work Presentations Homework Practical Sessions Exercises		
Assessment Methods Description (%)	Quizzes		25%
	Assignments		25%
	Final Exam		50%

Learning Outcomes	After completion of this course, students should be able to: - Establish knowledge and understanding of intermediate Python programming concepts - Analyze, design, code, compile, and debug programs in Python language. - Develop programs involving decision structures, loops, and functions. - Use different data types in a computer program		
Prerequisite Course(s)	-		
Language of Instruction	English Language		
Mandatory Literature	Tony Gaddis: Starting Out with Python, 5th edition		
Recommended Literature	-		
ECTS (ALLOCATED BASED ON STUDENT’S WORKLOAD)			
Activities	Količina	Trajanje	Opterećenje
Lecture (15 weeks x Lecture hours per week)	15	3	45
Laboratory / Practice (15 weeks x Laboratory / Practice hours per week)	15	2	30
Final Examination (1 week)	1	5	5
Preparation for Final Examination	1	60	60
Assignment / Homework / Project	10	3	30
Seminar / Presentation / Quizzes	10	3	30
Total Workload			200
ECTS Credit (Total Workload / 25)			8